# Approximating price series, part 3: MLP of t (tanh) and Fourier-feature MLP of t

Approximation study, 2020-01-02 to 2023-07-31 daily closes, three tickers. All numbers recomputed from the saved fits (approx_study/fits/*.json).

## 1. What is being measured

This is an in-sample APPROXIMATION task, not forecasting. Each model receives the entire close-price series of a stock, p(t), with t rescaled to [0, 1] (905 points for CBA, 901 for TSLA and LLY; prices in the file's whole-series Min-Max scale, 0 = lowest close of the period, 1 = highest), and must reproduce it as a function of t with a fixed number of parameters. The score is the RMSE and the maximum absolute error between the fitted curve and the actual series, and how both fall as the parameter budget grows (about 15, 30, 60, 120, 240). There is no train/test split and no prediction; in-sample fit says nothing about forecasting. For scale, the dotted line in the error plots is the one-day-ahead naive forecasting error of the same series (0.0201 CBA, 0.0255 TSLA, 0.0148 LLY).

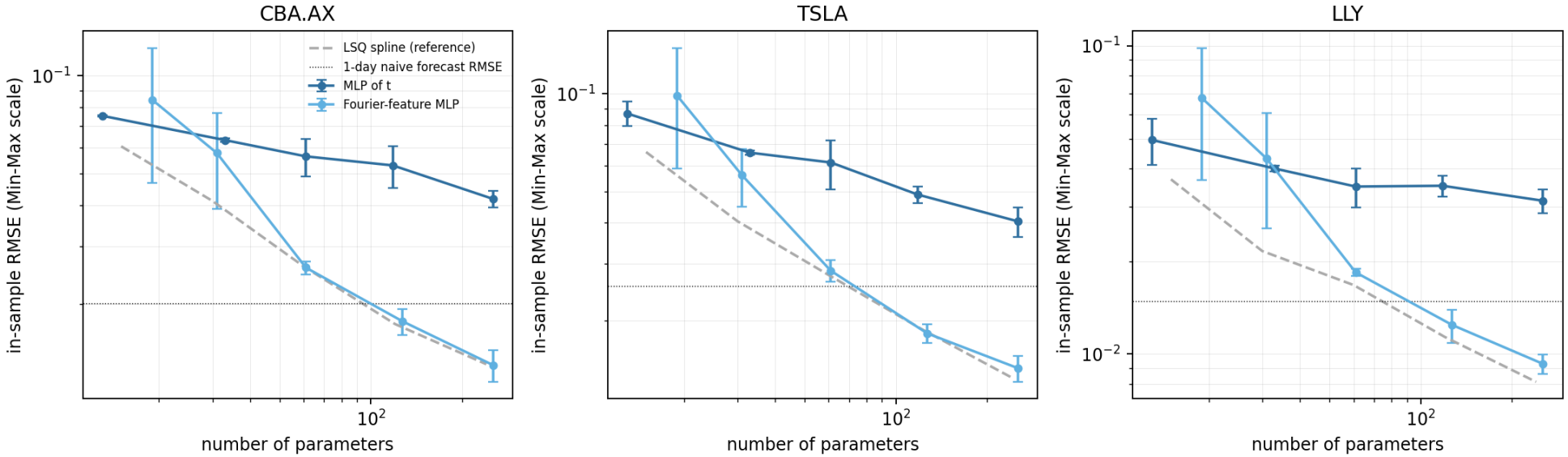
## 2. The two models in this document

**MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output**
**Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network**
• Both are PyTorch MLPs with two hidden layers of width w and tanh units, trained full-batch with Adam (lr 0.01, cosine schedule) for 20,000 steps, keeping the best-loss weights. Three seeds (0, 1, 2).
• MLP of t: input is the scalar t. Parameters = w^2 + 4w + 1.
• Fourier-feature MLP: the identical network, but its input is a fixed random map of t into 2 n_freq sinusoids, frequencies B ~ N(0, 6^2) drawn once per seed and NOT trained (Tancik et al. 2020). Parameters = w^2 + (2 n_freq + 3) w + 1; the width w is reduced so the total matches the budget.
• The one thing that changes is the input basis. Tancik et al. prove a plain MLP of a coordinate learns low frequencies first and high frequencies extremely slowly (spectral bias); the Fourier map removes that.
• Training length matters for the plain MLP: 4,000 steps gave RMSE 0.074 on CBA at 61 parameters, 20,000 steps 0.048; all results below use 20,000 steps.

## 3. Settings per budget

| target budget | 15 | 30 | 60 | 120 |
|---|---|---|---|---|
| MLP of t actual params | 13 | 33 | 61 | 118 |
| Fourier-feature MLP actual params | 19 | 31 | 61 | 127 |
| setting | MLP width w = 2; Fourier MLP (n_freq, w) = (2, 2) | MLP width w = 4; Fourier MLP (n_freq, w) = (2, 3) | MLP width w = 6; Fourier MLP (n_freq, w) = (4, 4) | MLP width w = 9; Fourier MLP (n_freq, w) |

## 4. Error versus number of parameters (all tickers)




Points are the mean over seeds, bars the sample standard deviation across seeds (absent for deterministic models). Log-log axes: a straight line means error falls as a power of the parameter count.

Slope of log RMSE against log parameters, budgets 30-240 (how fast each model improves with size):

| model | CBA.AX | TSLA | LLY | mean |
|---|---|---|---|---|
| MLP of t | -0.20 | -0.25 | -0.11 | -0.18 |
| Fourier-feature MLP | -0.69 | -0.65 | -0.71 | -0.68 |

# 5. Every fit: full table

One row per fit (ticker, model, budget, seed). RMSE and max |error| in the ticker's Min-Max scale; seconds = wall-clock fit time.

| ticker | model | budget | seed | params | RMSE | max \|err\| | seconds | setting |
|---|---|---|---|---|---|---|---|---|
| CBA.AX | mlp | 15 | 0 | 13 | 0.0758 | 0.2554 | 13.0 | w=2 |
| CBA.AX | mlp | 15 | 1 | 13 | 0.0748 | 0.2414 | 11.2 | w=2 |
| CBA.AX | mlp | 15 | 2 | 13 | 0.0752 | 0.2421 | 11.2 | w=2 |
| CBA.AX | mlp | 30 | 0 | 33 | 0.0637 | 0.2173 | 10.6 | w=4 |
| CBA.AX | mlp | 30 | 1 | 33 | 0.0641 | 0.2106 | 10.3 | w=4 |
| CBA.AX | mlp | 30 | 2 | 33 | 0.0624 | 0.1884 | 10.7 | w=4 |
| CBA.AX | mlp | 60 | 0 | 61 | 0.0482 | 0.1924 | 11.1 | w=6 |
| CBA.AX | mlp | 60 | 1 | 61 | 0.0626 | 0.2160 | 11.9 | w=6 |
| CBA.AX | mlp | 60 | 2 | 61 | 0.0588 | 0.2011 | 12.0 | w=6 |
| CBA.AX | mlp | 120 | 0 | 118 | 0.0493 | 0.1958 | 10.5 | w=9 |
| CBA.AX | mlp | 120 | 1 | 118 | 0.0478 | 0.1977 | 11.1 | w=9 |
| CBA.AX | mlp | 120 | 2 | 118 | 0.0621 | 0.2114 | 11.4 | w=9 |
| CBA.AX | mlp | 240 | 0 | 253 | 0.0393 | 0.1297 | 12.3 | w=14 |
| CBA.AX | mlp | 240 | 1 | 253 | 0.0440 | 0.1618 | 14.7 | w=14 |
| CBA.AX | mlp | 240 | 2 | 253 | 0.0424 | 0.1403 | 12.9 | w=14 |
| CBA.AX | fmlp | 15 | 0 | 19 | 0.0660 | 0.2293 | 10.9 | n_freq=2, w=2, sigma=6 |
| CBA.AX | fmlp | 15 | 1 | 19 | 0.1272 | 0.6627 | 10.9 | n_freq=2, w=2, sigma=6 |
| CBA.AX | fmlp | 15 | 2 | 19 | 0.0595 | 0.1903 | 10.2 | n_freq=2, w=2, sigma=6 |
| CBA.AX | fmlp | 30 | 0 | 31 | 0.0544 | 0.2045 | 10.4 | n_freq=2, w=3, sigma=6 |
| CBA.AX | fmlp | 30 | 1 | 31 | 0.0783 | 0.3452 | 10.4 | n_freq=2, w=3, sigma=6 |
| CBA.AX | fmlp | 30 | 2 | 31 | 0.0411 | 0.1372 | 10.7 | n_freq=2, w=3, sigma=6 |
| CBA.AX | fmlp | 60 | 0 | 61 | 0.0271 | 0.1054 | 14.0 | n_freq=4, w=4, sigma=6 |
| CBA.AX | fmlp | 60 | 1 | 61 | 0.0249 | 0.0793 | 10.5 | n_freq=4, w=4, sigma=6 |
| CBA.AX | fmlp | 60 | 2 | 61 | 0.0254 | 0.1052 | 11.5 | n_freq=4, w=4, sigma=6 |
| CBA.AX | fmlp | 120 | 0 | 127 | 0.0195 | 0.0909 | 10.5 | n_freq=6, w=6, sigma=6 |
| CBA.AX | fmlp | 120 | 1 | 127 | 0.0170 | 0.0800 | 10.5 | n_freq=6, w=6, sigma=6 |

| ticker | model | budget | seed | params | RMSE | max \|err\| | seconds | setting |
|---|---|---|---|---|---|---|---|---|
| CBA.AX | fmlp | 120 | 2 | 127 | 0.0165 | 0.0781 | 11.1 | n_freq=6, w=6, sigma=6 |
| CBA.AX | fmlp | 240 | 0 | 253 | 0.0128 | 0.0615 | 12.1 | n_freq=8, w=9, sigma=6 |
| CBA.AX | fmlp | 240 | 1 | 253 | 0.0145 | 0.0751 | 14.1 | n_freq=8, w=9, sigma=6 |
| CBA.AX | fmlp | 240 | 2 | 253 | 0.0116 | 0.0646 | 13.5 | n_freq=8, w=9, sigma=6 |
| TSLA | mlp | 15 | 0 | 13 | 0.0930 | 0.2976 | 11.8 | w=2 |
| TSLA | mlp | 15 | 1 | 13 | 0.0785 | 0.3023 | 11.6 | w=2 |
| TSLA | mlp | 15 | 2 | 13 | 0.0890 | 0.3080 | 11.3 | w=2 |
| TSLA | mlp | 30 | 0 | 33 | 0.0650 | 0.2473 | 11.8 | w=4 |
| TSLA | mlp | 30 | 1 | 33 | 0.0654 | 0.2448 | 11.9 | w=4 |
| TSLA | mlp | 30 | 2 | 33 | 0.0669 | 0.2549 | 11.4 | w=4 |
| TSLA | mlp | 60 | 0 | 61 | 0.0530 | 0.2393 | 11.0 | w=6 |
| TSLA | mlp | 60 | 1 | 61 | 0.0732 | 0.2521 | 10.3 | w=6 |
| TSLA | mlp | 60 | 2 | 61 | 0.0578 | 0.2453 | 10.3 | w=6 |
| TSLA | mlp | 120 | 0 | 118 | 0.0457 | 0.1785 | 10.5 | w=9 |
| TSLA | mlp | 120 | 1 | 118 | 0.0510 | 0.2378 | 11.2 | w=9 |
| TSLA | mlp | 120 | 2 | 118 | 0.0499 | 0.1982 | 11.2 | w=9 |
| TSLA | mlp | 240 | 0 | 253 | 0.0356 | 0.1480 | 11.7 | w=14 |
| TSLA | mlp | 240 | 1 | 253 | 0.0433 | 0.1695 | 11.4 | w=14 |
| TSLA | mlp | 240 | 2 | 253 | 0.0422 | 0.1510 | 11.5 | w=14 |
| TSLA | fmlp | 15 | 0 | 19 | 0.0700 | 0.2259 | 11.3 | n_freq=2, w=2, sigma=6 |
| TSLA | fmlp | 15 | 1 | 19 | 0.1440 | 0.5914 | 11.4 | n_freq=2, w=2, sigma=6 |
| TSLA | fmlp | 15 | 2 | 19 | 0.0816 | 0.3109 | 11.7 | n_freq=2, w=2, sigma=6 |
| TSLA | fmlp | 30 | 0 | 31 | 0.0550 | 0.2344 | 11.8 | n_freq=2, w=3, sigma=6 |
| TSLA | fmlp | 30 | 1 | 31 | 0.0681 | 0.3133 | 14.2 | n_freq=2, w=3, sigma=6 |
| TSLA | fmlp | 30 | 2 | 31 | 0.0454 | 0.1924 | 10.4 | n_freq=2, w=3, sigma=6 |
| TSLA | fmlp | 60 | 0 | 61 | 0.0289 | 0.1162 | 10.5 | n_freq=4, w=4, sigma=6 |
| TSLA | fmlp | 60 | 1 | 61 | 0.0304 | 0.1423 | 10.2 | n_freq=4, w=4, sigma=6 |
| TSLA | fmlp | 60 | 2 | 61 | 0.0261 | 0.1146 | 10.1 | n_freq=4, w=4, sigma=6 |

| ticker | model | budget | seed | params | RMSE | max \|err\| | seconds | setting |
|---|---|---|---|---|---|---|---|---|
| TSLA | fmlp | 120 | 0 | 127 | 0.0195 | 0.0862 | 10.4 | n_freq=6, w=6, sigma=6 |
| TSLA | fmlp | 120 | 1 | 127 | 0.0181 | 0.0745 | 10.7 | n_freq=6, w=6, sigma=6 |
| TSLA | fmlp | 120 | 2 | 127 | 0.0171 | 0.0829 | 12.9 | n_freq=6, w=6, sigma=6 |
| TSLA | fmlp | 240 | 0 | 253 | 0.0151 | 0.0807 | 11.4 | n_freq=8, w=9, sigma=6 |
| TSLA | fmlp | 240 | 1 | 253 | 0.0150 | 0.0724 | 10.6 | n_freq=8, w=9, sigma=6 |
| TSLA | fmlp | 240 | 2 | 253 | 0.0127 | 0.0497 | 11.0 | n_freq=8, w=9, sigma=6 |
| LLY | mlp | 15 | 0 | 13 | 0.0586 | 0.2014 | 10.6 | w=2 |
| LLY | mlp | 15 | 1 | 13 | 0.0421 | 0.1073 | 10.5 | w=2 |
| LLY | mlp | 15 | 2 | 13 | 0.0478 | 0.1351 | 10.4 | w=2 |
| LLY | mlp | 30 | 0 | 33 | 0.0408 | 0.1135 | 10.4 | w=4 |
| LLY | mlp | 30 | 1 | 33 | 0.0391 | 0.1042 | 10.3 | w=4 |
| LLY | mlp | 30 | 2 | 33 | 0.0400 | 0.1114 | 10.7 | w=4 |
| LLY | mlp | 60 | 0 | 61 | 0.0379 | 0.1113 | 10.9 | w=6 |
| LLY | mlp | 60 | 1 | 61 | 0.0379 | 0.1107 | 11.2 | w=6 |
| LLY | mlp | 60 | 2 | 61 | 0.0291 | 0.0870 | 10.4 | w=6 |
| LLY | mlp | 120 | 0 | 118 | 0.0343 | 0.1130 | 10.9 | w=9 |
| LLY | mlp | 120 | 1 | 118 | 0.0382 | 0.1128 | 10.8 | w=9 |
| LLY | mlp | 120 | 2 | 118 | 0.0330 | 0.0991 | 11.1 | w=9 |
| LLY | mlp | 240 | 0 | 253 | 0.0302 | 0.1052 | 11.6 | w=14 |
| LLY | mlp | 240 | 1 | 253 | 0.0347 | 0.1120 | 11.3 | w=14 |
| LLY | mlp | 240 | 2 | 253 | 0.0294 | 0.0907 | 11.3 | w=14 |
| LLY | fmlp | 15 | 0 | 19 | 0.0388 | 0.1119 | 10.5 | n_freq=2, w=2, sigma=6 |
| LLY | fmlp | 15 | 1 | 19 | 0.1002 | 0.4593 | 10.5 | n_freq=2, w=2, sigma=6 |
| LLY | fmlp | 15 | 2 | 19 | 0.0639 | 0.2470 | 10.2 | n_freq=2, w=2, sigma=6 |
| LLY | fmlp | 30 | 0 | 31 | 0.0357 | 0.1160 | 12.0 | n_freq=2, w=3, sigma=6 |
| LLY | fmlp | 30 | 1 | 31 | 0.0631 | 0.2697 | 10.1 | n_freq=2, w=3, sigma=6 |
| LLY | fmlp | 30 | 2 | 31 | 0.0305 | 0.1091 | 10.6 | n_freq=2, w=3, sigma=6 |
| LLY | fmlp | 60 | 0 | 61 | 0.0189 | 0.0694 | 10.6 | n_freq=4, w=4, sigma=6 |

| ticker | model | budget | seed | params | RMSE | max \|err\| | seconds | setting |
|---|---|---|---|---|---|---|---|---|
| LLY | fmlp | 60 | 1 | 61 | 0.0183 | 0.0767 | 10.6 | n_freq=4, w=4, sigma=6 |
| LLY | fmlp | 60 | 2 | 61 | 0.0180 | 0.0694 | 10.2 | n_freq=4, w=4, sigma=6 |
| LLY | fmlp | 120 | 0 | 127 | 0.0139 | 0.0529 | 10.4 | n_freq=6, w=6, sigma=6 |
| LLY | fmlp | 120 | 1 | 127 | 0.0124 | 0.0516 | 12.6 | n_freq=6, w=6, sigma=6 |
| LLY | fmlp | 120 | 2 | 127 | 0.0109 | 0.0480 | 10.6 | n_freq=6, w=6, sigma=6 |
| LLY | fmlp | 240 | 0 | 253 | 0.0097 | 0.0473 | 11.0 | n_freq=8, w=9, sigma=6 |
| LLY | fmlp | 240 | 1 | 253 | 0.0097 | 0.0416 | 10.8 | n_freq=8, w=9, sigma=6 |
| LLY | fmlp | 240 | 2 | 253 | 0.0085 | 0.0378 | 10.6 | n_freq=8, w=9, sigma=6 |

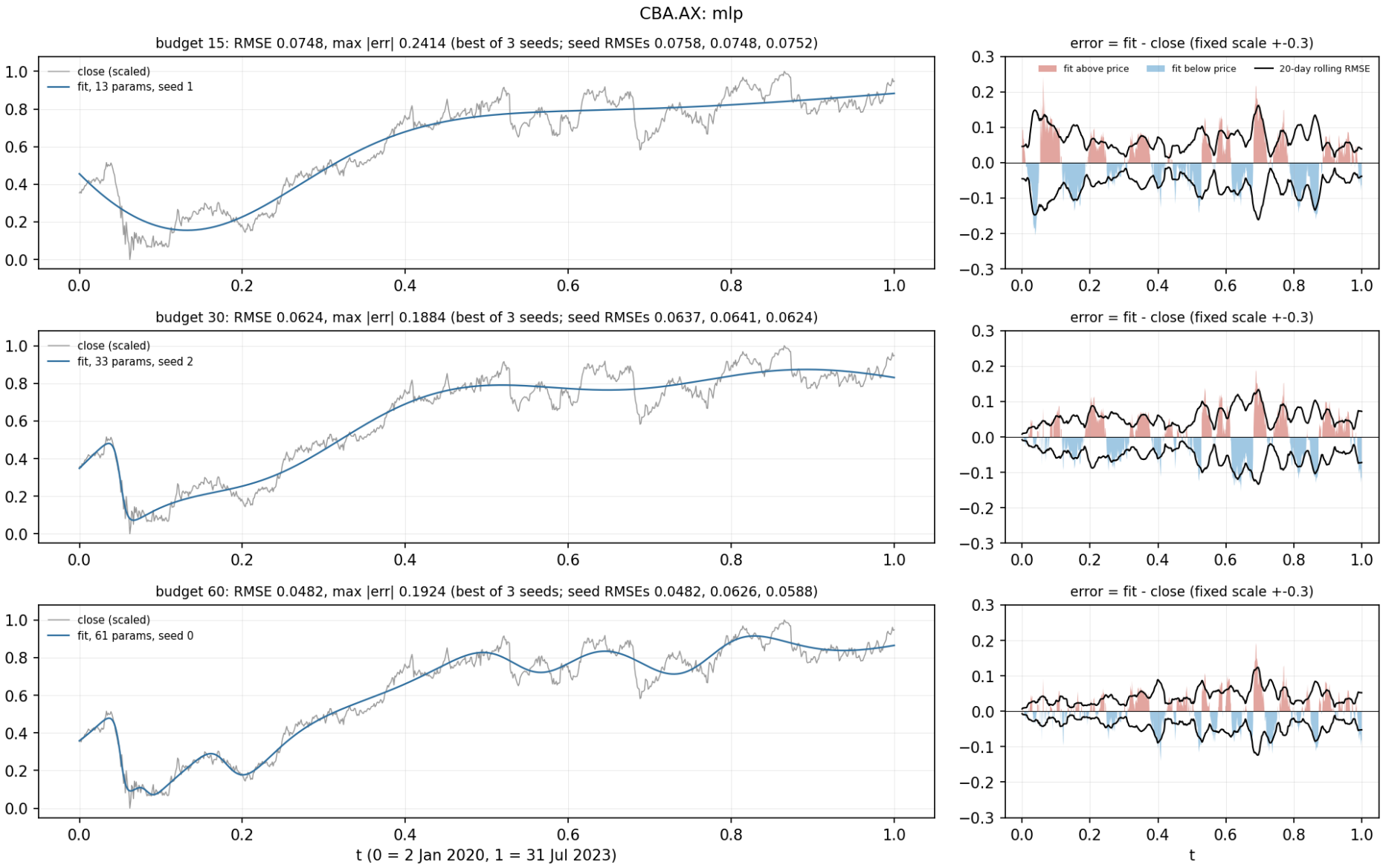
## 6. Summary per ticker and budget (mean +- sd over seeds)

| ticker | budget | MLP of t RMSE | max \|err\| | params | Fourier-feature MLP RMSE | max \|err\| | params |
|---|---|---|---|---|---|---|---|
| CBA.AX | 15 | 0.0753 +- 0.0005 | 0.2463 | 13 | 0.0842 +- 0.0373 | 0.3608 | 19 |
| CBA.AX | 30 | 0.0634 +- 0.0009 | 0.2054 | 33 | 0.0579 +- 0.0189 | 0.2290 | 31 |
| CBA.AX | 60 | 0.0566 +- 0.0074 | 0.2032 | 61 | 0.0258 +- 0.0012 | 0.0966 | 61 |
| CBA.AX | 120 | 0.0531 +- 0.0078 | 0.2016 | 118 | 0.0177 +- 0.0016 | 0.0830 | 127 |
| CBA.AX | 240 | 0.0419 +- 0.0024 | 0.1439 | 253 | 0.0130 +- 0.0014 | 0.0670 | 253 |
| TSLA | 15 | 0.0868 +- 0.0075 | 0.3026 | 13 | 0.0985 +- 0.0398 | 0.3760 | 19 |
| TSLA | 30 | 0.0658 +- 0.0010 | 0.2490 | 33 | 0.0562 +- 0.0114 | 0.2467 | 31 |
| TSLA | 60 | 0.0614 +- 0.0106 | 0.2456 | 61 | 0.0285 +- 0.0022 | 0.1244 | 61 |
| TSLA | 120 | 0.0489 +- 0.0028 | 0.2049 | 118 | 0.0182 +- 0.0012 | 0.0812 | 127 |
| TSLA | 240 | 0.0404 +- 0.0041 | 0.1562 | 253 | 0.0143 +- 0.0013 | 0.0676 | 253 |
| LLY | 15 | 0.0495 +- 0.0084 | 0.1479 | 13 | 0.0676 +- 0.0309 | 0.2728 | 19 |
| LLY | 30 | 0.0400 +- 0.0008 | 0.1097 | 33 | 0.0431 +- 0.0175 | 0.1649 | 31 |
| LLY | 60 | 0.0349 +- 0.0051 | 0.1030 | 61 | 0.0184 +- 0.0005 | 0.0718 | 61 |
| LLY | 120 | 0.0352 +- 0.0027 | 0.1083 | 118 | 0.0124 +- 0.0015 | 0.0508 | 127 |
| LLY | 240 | 0.0314 +- 0.0028 | 0.1026 | 253 | 0.0093 +- 0.0007 | 0.0422 | 253 |

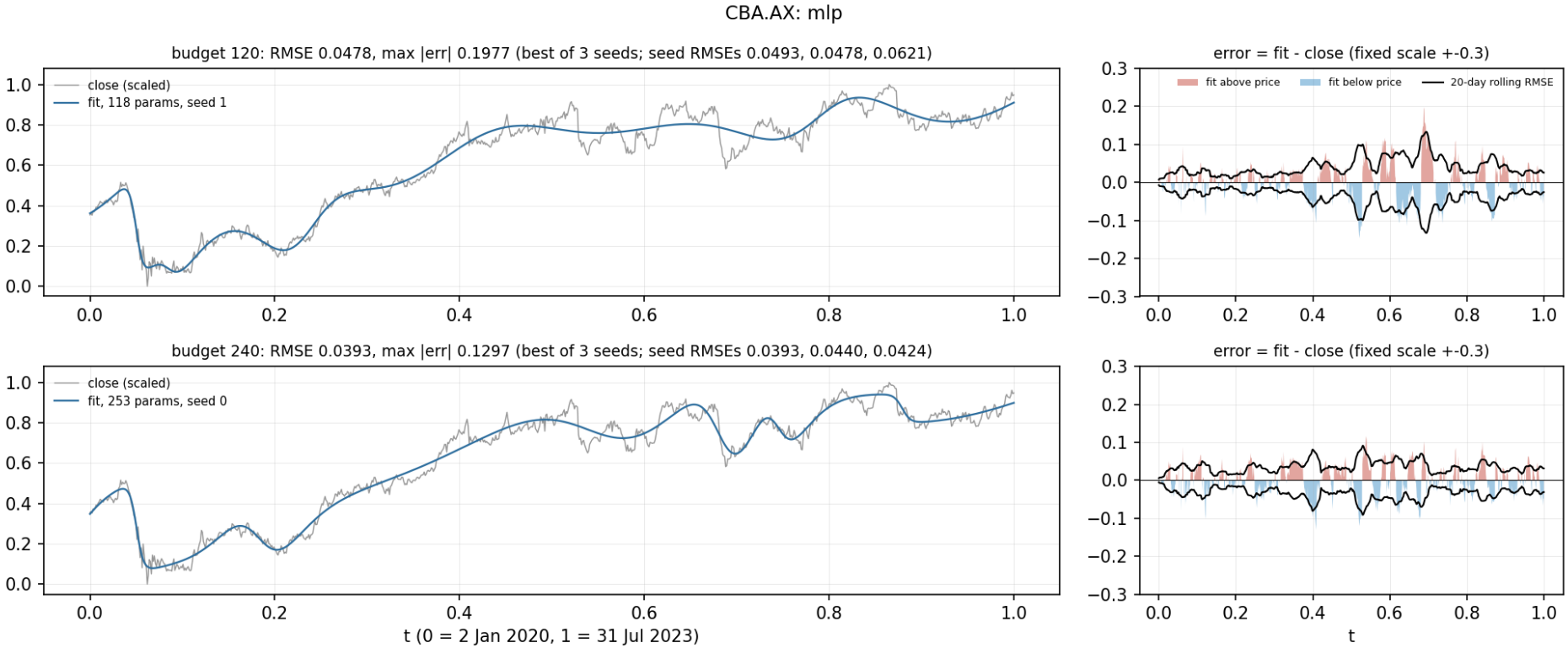
## 7. The fits themselves (following pages)

For every ticker, model and budget: left, the fitted curve over the actual series (best seed where there are seeds; all seed RMSEs in the title); right, the error fit - close on a fixed +-0.3 scale so panels are comparable: red shading where the fit sits above the price, blue where below, and the black envelope is the 20-day rolling RMSE (+-), i.e. the local size of the error with the day-to-day jitter smoothed out. Vertical lines in spline panels are knot positions.

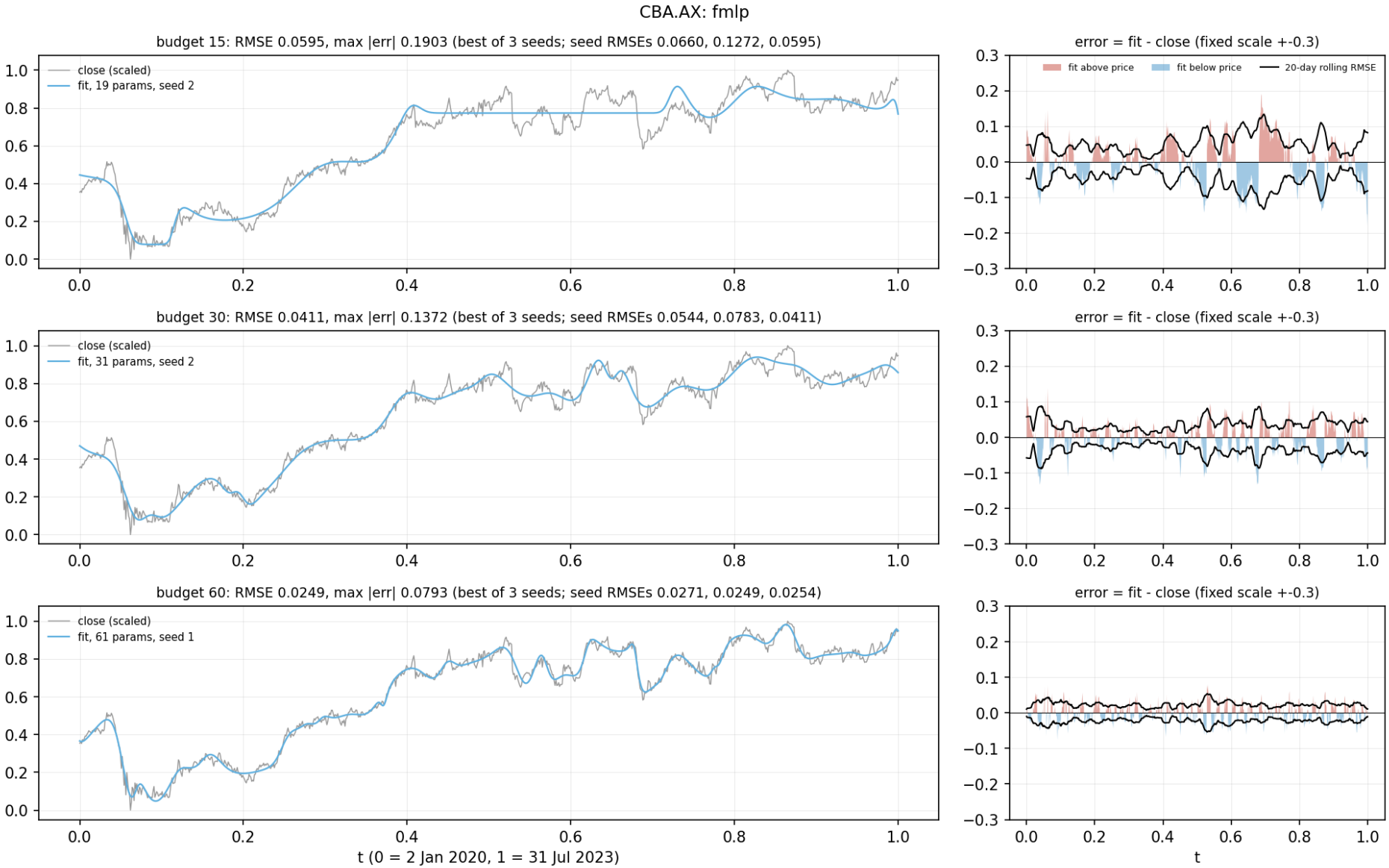
***CBA.AX (Commonwealth Bank, benchmark series), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 15, 30, 60)***



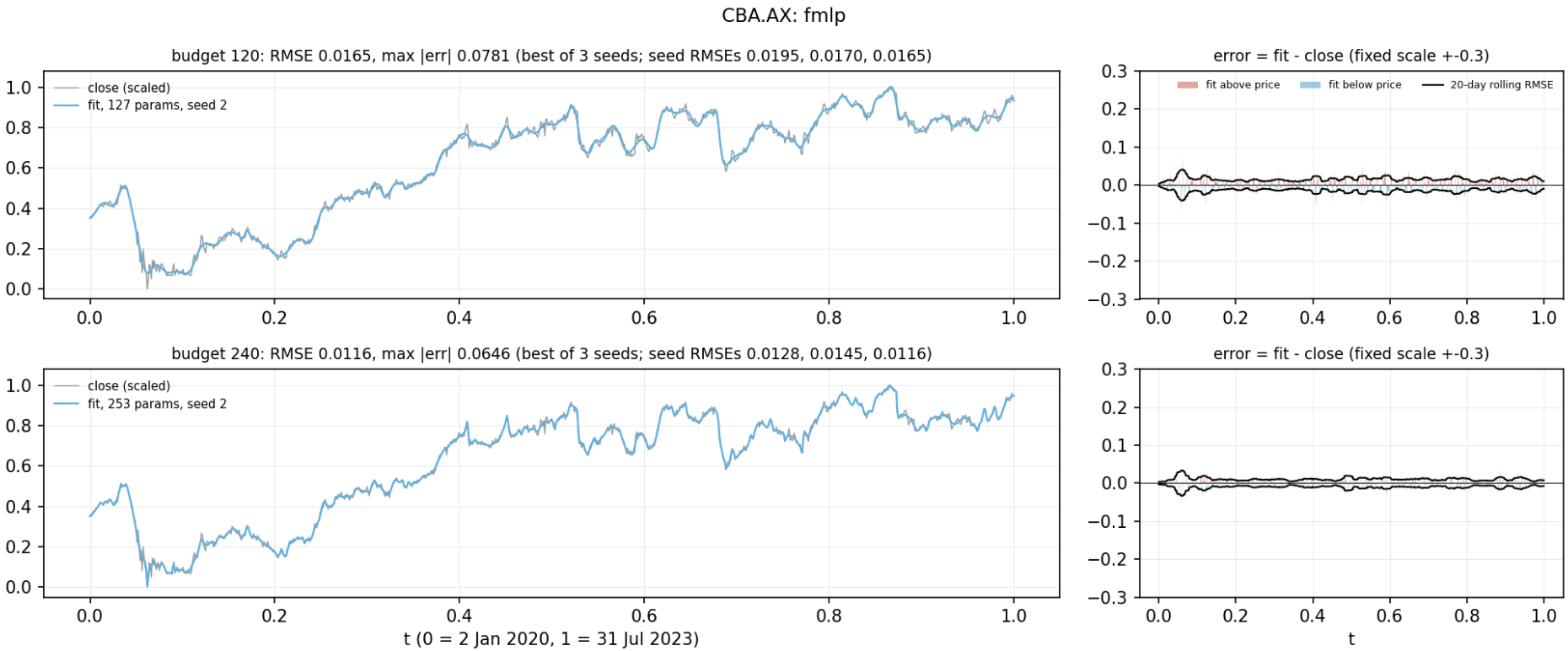
***CBA.AX (Commonwealth Bank, benchmark series), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 120, 240)***



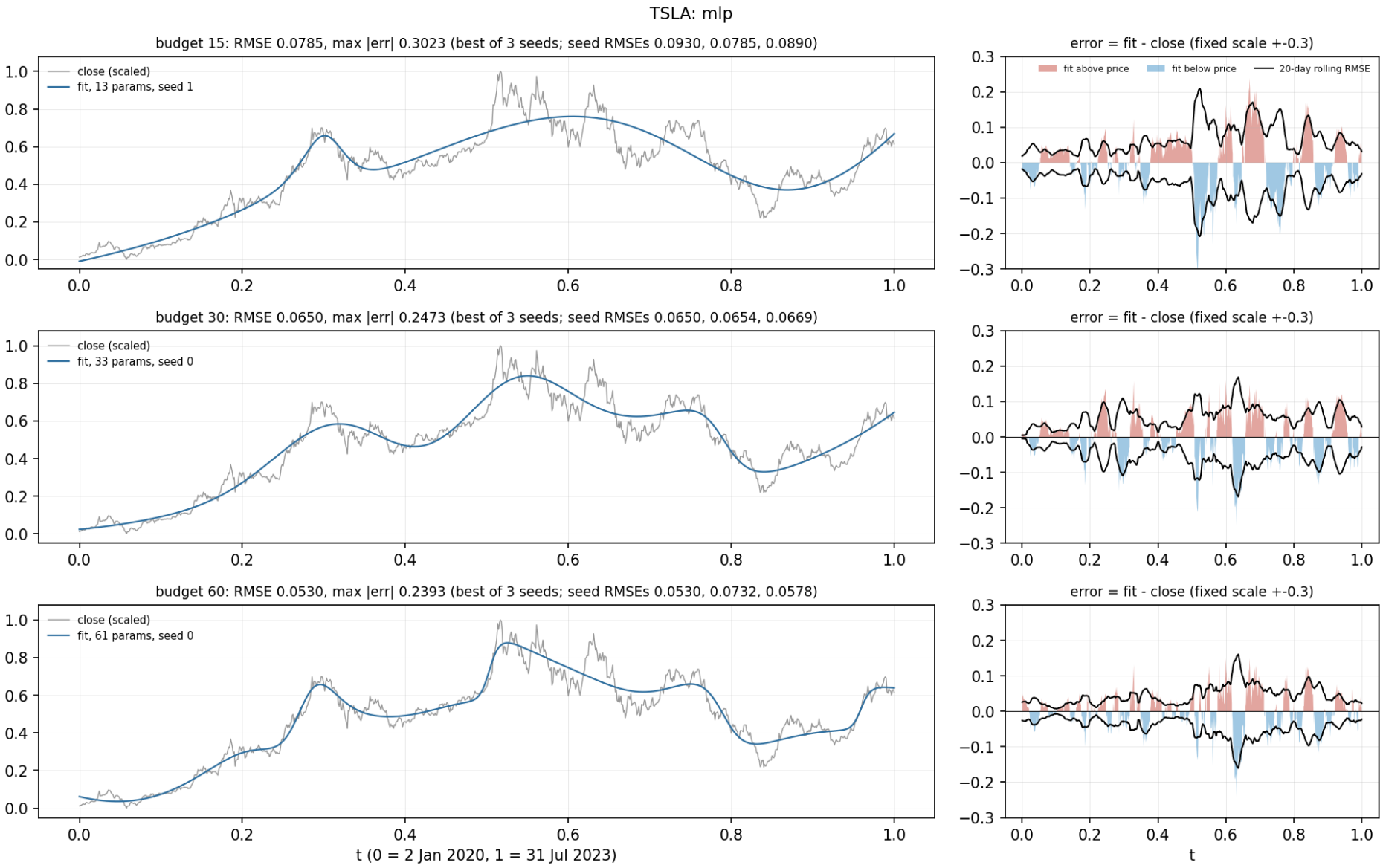
***CBA.AX (Commonwealth Bank, benchmark series), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 15, 30, 60)***



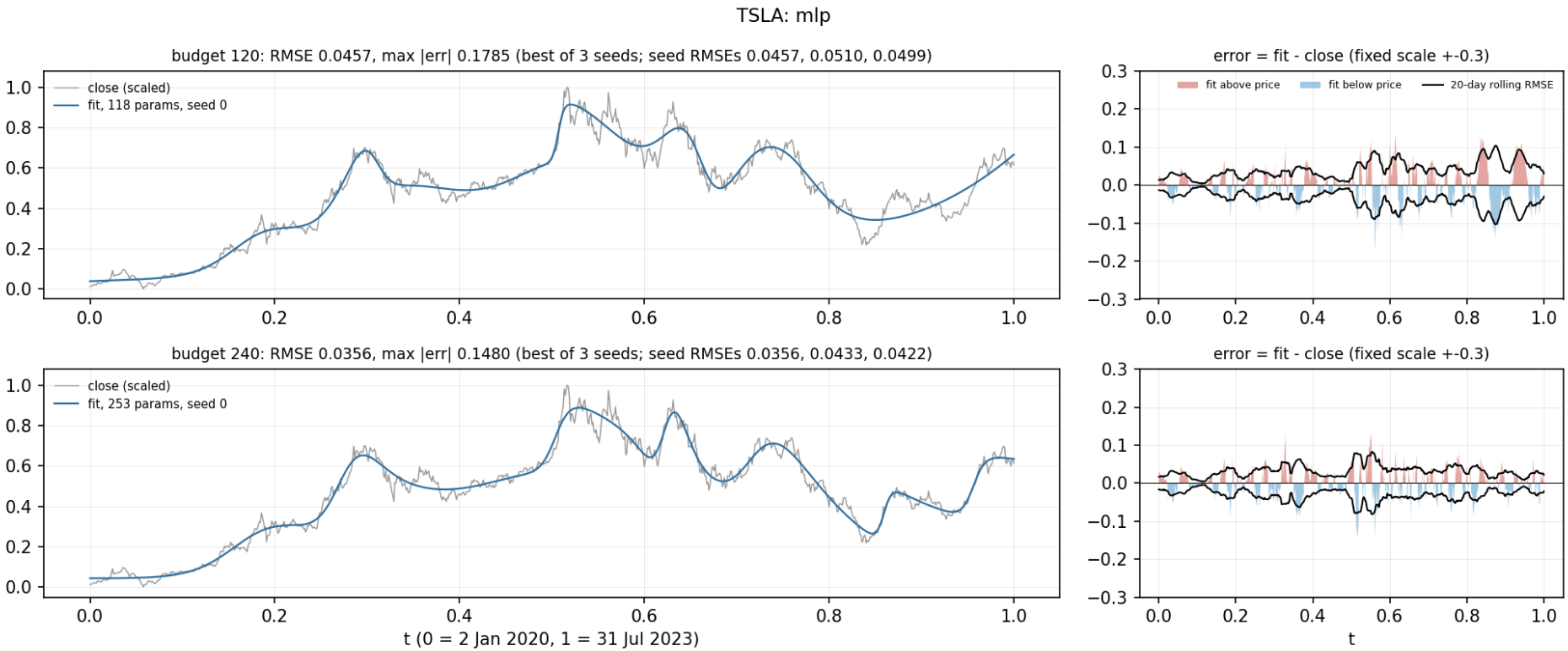
***CBA.AX (Commonwealth Bank, benchmark series), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 120, 240)***



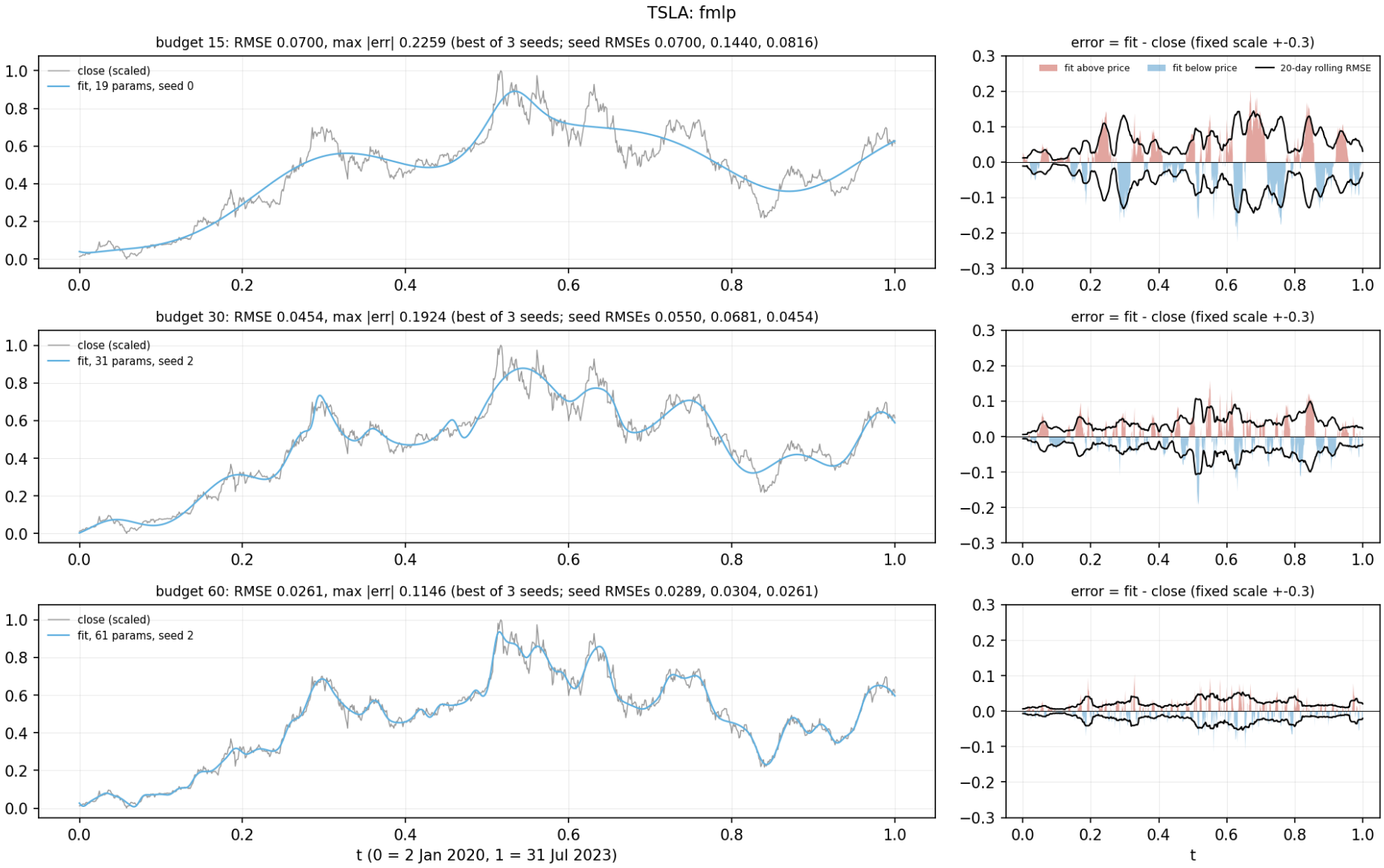
## *TSLA (Tesla, volatile), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 15, 30, 60)*



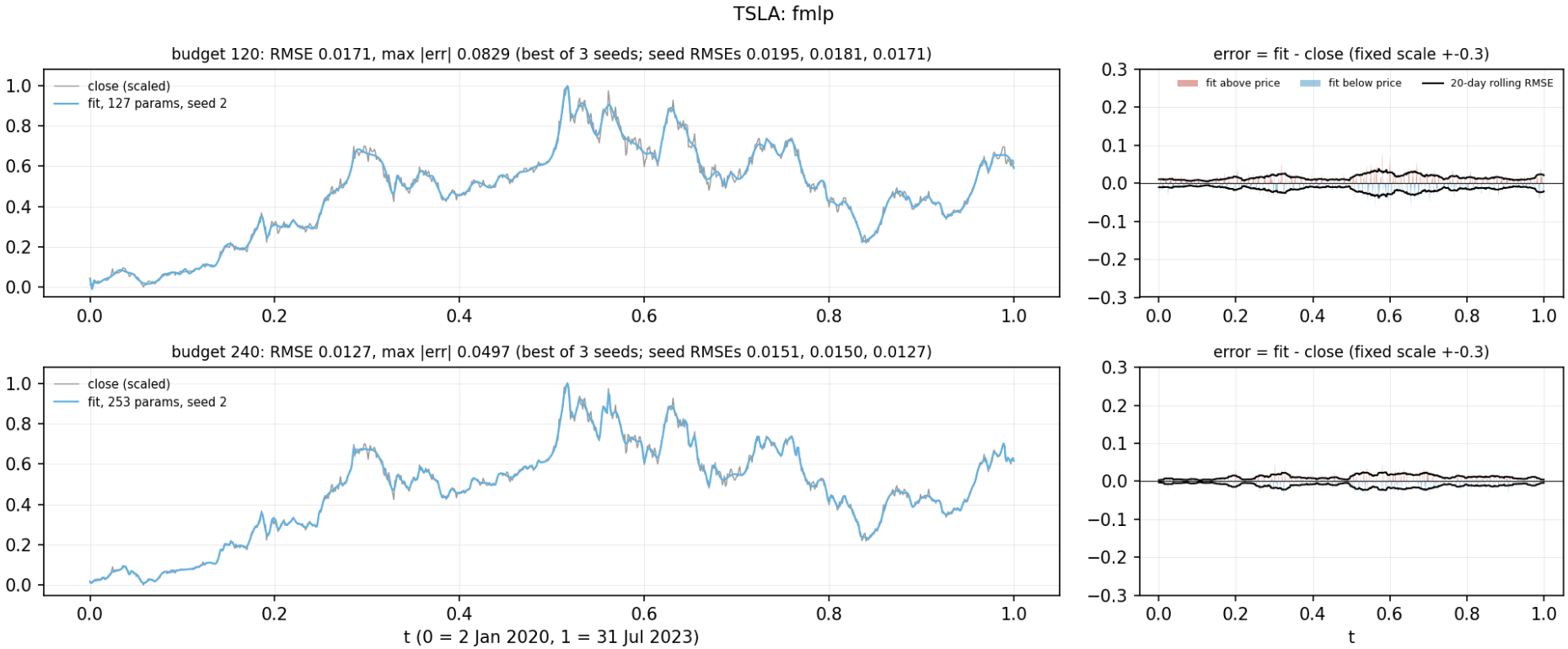
***TSLA (Tesla, volatile), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 120, 240)***



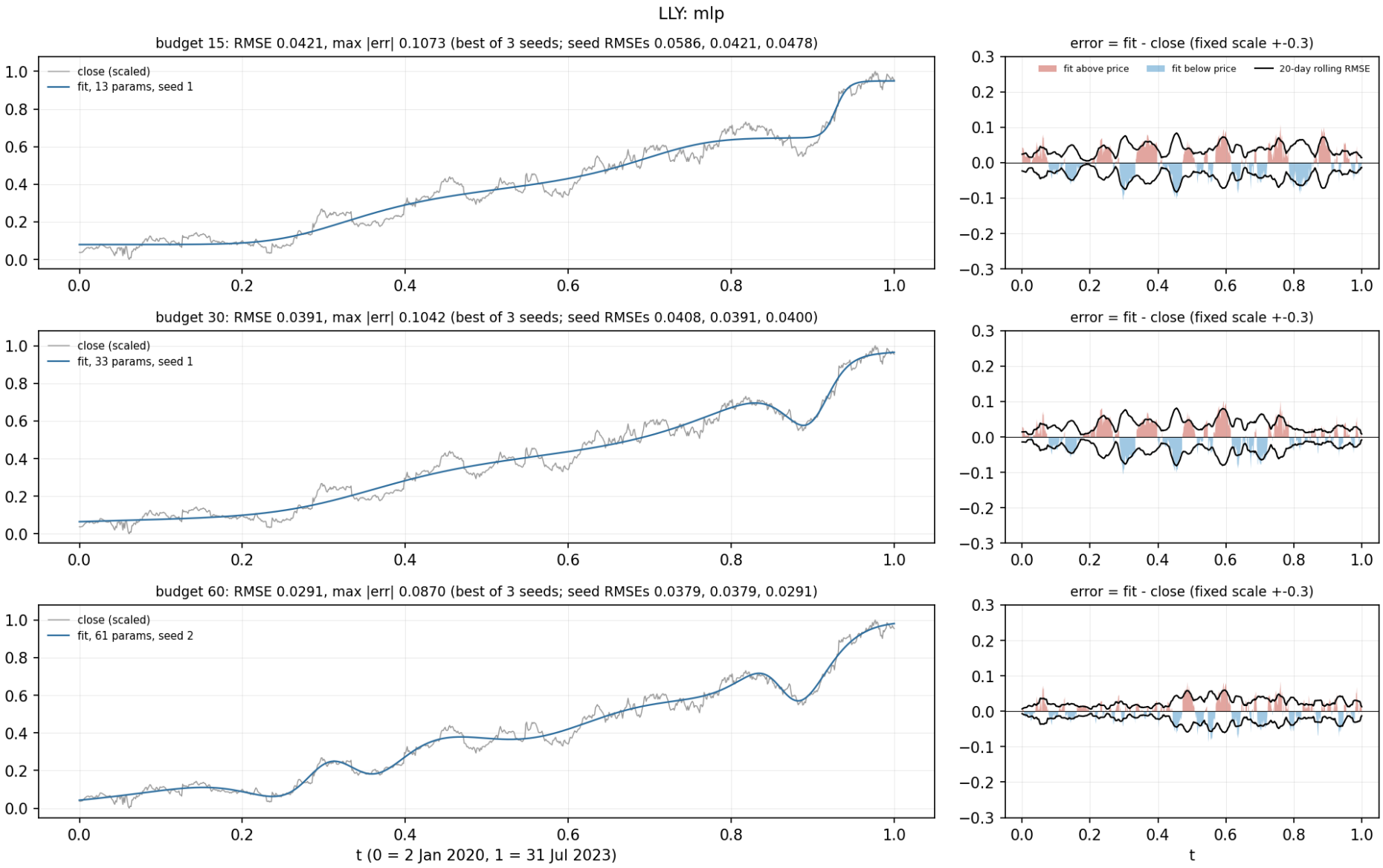
***TSLA (Tesla, volatile), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 15, 30, 60)***



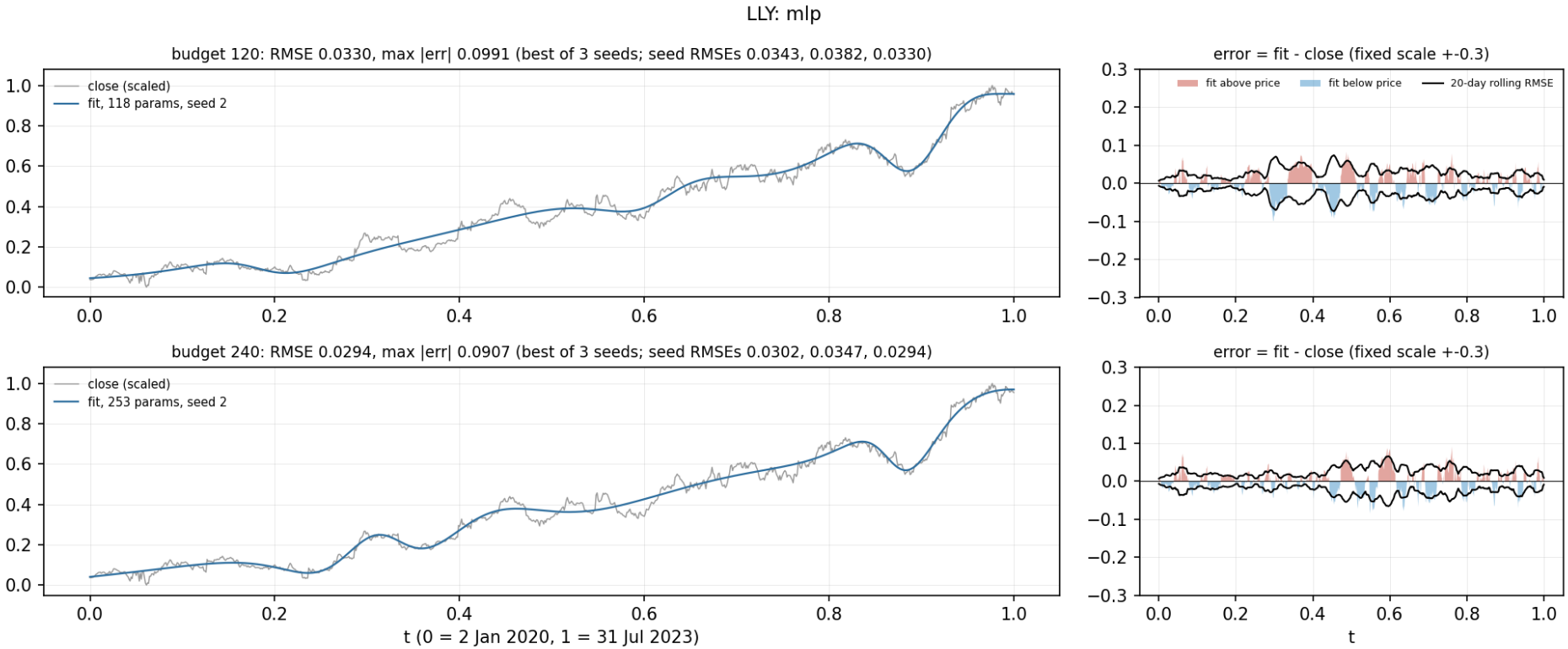
***TSLA (Tesla, volatile), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 120, 240)***



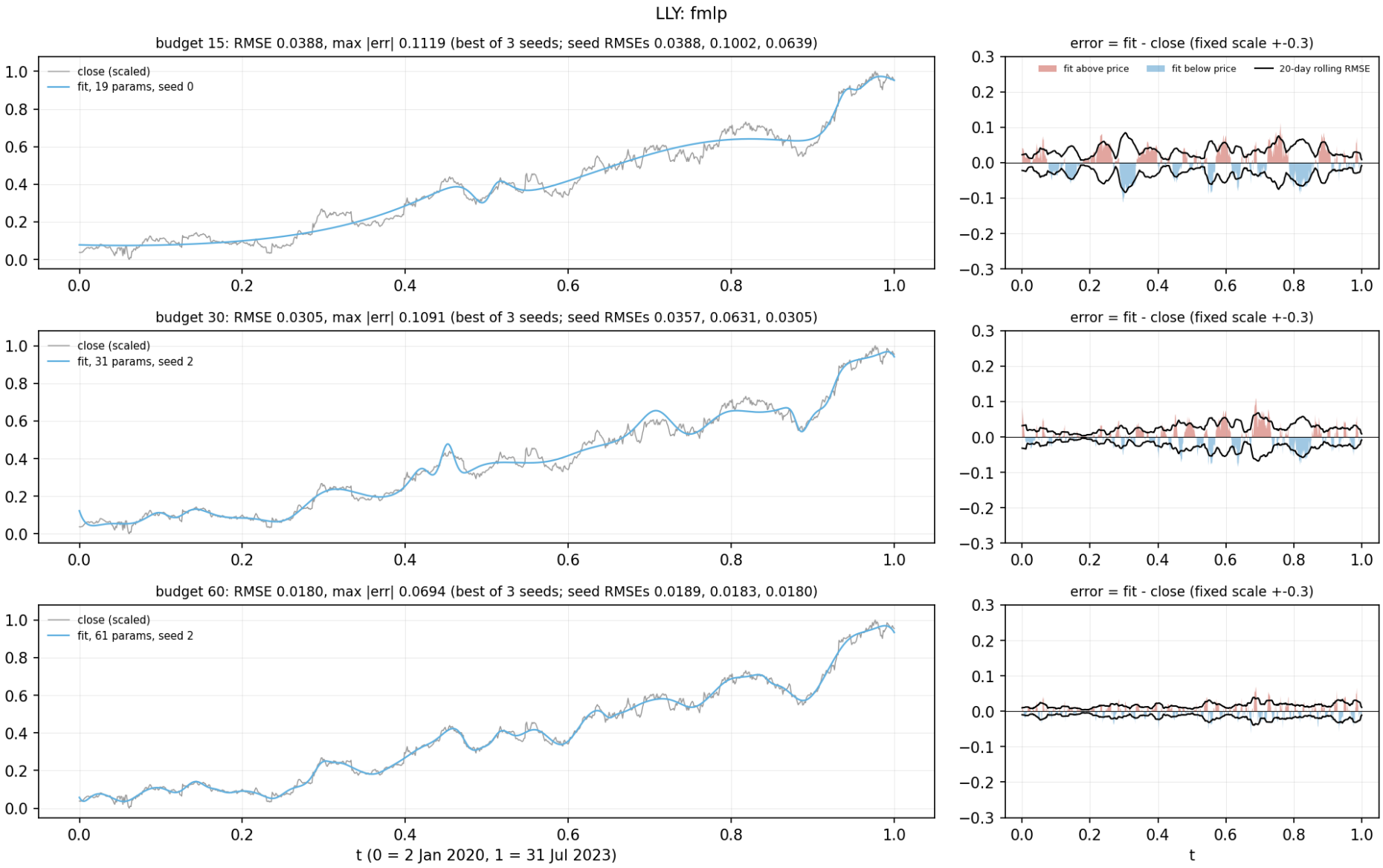
***LLY (Eli Lilly, stable trend), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 15, 30, 60)***



***LLY (Eli Lilly, stable trend), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 120, 240)***



***LLY (Eli Lilly, stable trend), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 15, 30, 60)***



***LLY (Eli Lilly, stable trend), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 120, 240)***

